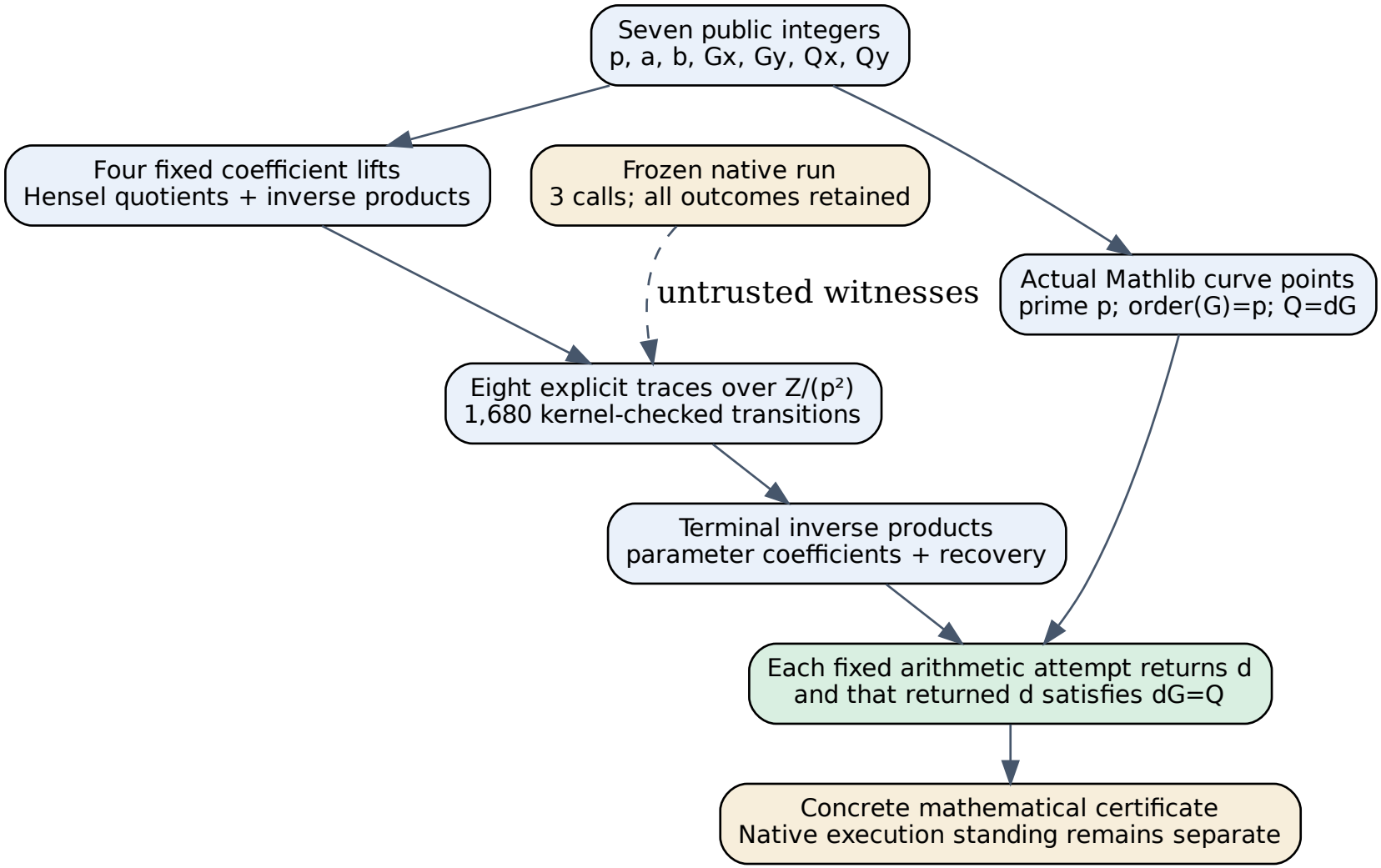
Seven public integers
p, a, b, Gx, Gy, Qx, Qy
Four fixed coefficient lifts
Hensel quotients + inverse products
Frozen native run
3 calls; all outcomes retained
untrusted witnesses
Actual Mathlib curve points
prime p; order(G)=p; Q=dG
Eight explicit traces over Z/(p²)
1,680 kernel-checked transitions
Terminal inverse products
parameter coefficients + recovery
Each fixed arithmetic attempt returns d
and that returned d satisfies dG=Q
Concrete mathematical certificate
Native execution standing remains separate